Ricardo (Zhicong) Xie

MECHANICAL ENGINEERING | PROTOTYPING, TESTING AND SIMULATION
Auckland, NZ | +64 29 027 29390 | zxie588@aucklanduni.ac.nz | zxie588-alt.github.io | linkedin.com/in/ricardo-xie-6394643b4

PROFILE

- Mechanical engineering master's student who builds and proves real hardware: designed, fabricated, wired and power-tested a wearable energy-harvesting backpack that beat its 30–50 mW target, and uses CFD (Fluent) and FEA (Abaqus) to check designs.
- **Available full-time late November 2026 – late February 2027** (University of Auckland summer break).

EDUCATION

University of Auckland | Master of Engineering Studies (Mechanical) Feb 2026 – Jul 2027 (expected)
GPA 6.0/9.0 | Research Methods (A), Advanced Computational Fluid Dynamics (A-), Composite Materials (B)

Huazhong Agricultural University | BEng Mechanical Engineering (Mechatronics) 2021 – 2025
Mechanical design, electrical and electronic engineering, sensors, embedded control, mechatronic integration

PROJECTS

Energy-Harvesting Backpack Prototype | Lihua Tang Group, University of Auckland Jul – Oct 2026

- **Design:** modelled and designed the suspended-load platform in SolidWorks: twin guide rods, parallel elastic bands and a rack-and-pinion drive that turns **±35 mm** of payload travel into generator rotation (70 mm stroke ≈ 1 rev), designed for payloads of tens of kg. A fixed hole pattern lets the drive be upgraded without recutting the plates.
- **Build:** laser-cut the acrylic plates, 3D-printed parts, welded the generator support structure and assembled the platform; found a printed-hole/M5 clearance problem, corrected the tolerance and updated the CAD.
- **Wiring and fault-finding:** wired every coil of the group's generator prototype to a single output. When it gave no output, checked continuity joint by joint with a multimeter and traced the fault to a **coil lead worn through by friction** on an internal contact surface; repaired it and restored output.
- **Power testing (my own):** set up a 3.3 kΩ load and oscilloscope capture. Walking at 4 km/h with the backpack on, it delivered **55 mW average, beating the 30–50 mW project target** (81 mW hand-cranked; about 205 mW during vigorous movement at 5 km/h). Calculated RMS power point by point from raw samples. Earlier took part in doctoral-led shaker tests and phone-charging and pulse-oximeter demos.

Solar Chimney Natural Convection and Conjugate Heat Transfer | ANSYS Fluent / CFX Course research project

- Built a 2D axisymmetric three-region conjugate heat transfer model of buoyancy-driven flow; mesh independence across three grids (36,847-cell production mesh, 0.90% outlet-velocity difference to the fine mesh).
- Found that the reference paper's 12.2 m/s draught would need a ~299 K temperature rise, inconsistent with its reported 42–46 °C, and treated it as an uncertain benchmark in the technical report.

Composite Formula SAE Front Wing Structural Analysis | Abaqus Course project

- S4R shell and composite layup model under aerodynamic, lateral and 4.1 g bump cases; identified the governing case with ~74% peak stress utilisation.

Automated Dynamic Weighing of Pig Carcasses | STM32, HX711 Sep 2022 – Jan 2024

- Developed load-cell acquisition, embedded logic and serial/Bluetooth transfer for weighing pig carcasses in motion on a processing line; the project software is registered as Software Copyright No. 2024SR0141888 (holder: Huazhong Agricultural University).

SKILLS

- **CAD / CAE:** SolidWorks (proficient: parts, assemblies, drawings); ANSYS Fluent/CFX, Abaqus, AutoCAD/DXF (working); MATLAB/Simulink (basic)
- **Programming:** Python (real-time device-control IoT scripts in a UoA postgraduate course, data processing, small automation tools); embedded C on STM32 (basic)
- **Hands-on:** 3D printing, laser cutting, CNC training, arc welding and electronics soldering, multimeter fault-finding, mechanical assembly and bench testing
- **Languages:** English (IELTS 7.0; CET-6), Mandarin (native)

LEADERSHIP AND COMMUNITY

- **Workplace Insights Programme**, UoA Career Development and Employability Services (Aug – Oct 2026, completed): workshops on NZ workplaces and networking; catch-ups with a University of Auckland alumni volunteer
- **Photography Director**, Chinese Postgraduate Society (CNPG), New Zealand (2026 – present)
- **Head of Student News Department**, Huazhong Agricultural University: led news writing and photography, task allocation and deadlines
- **Volunteer**, Auckland Library of Tools (2026 – present); AUSA orientation volunteer